THE STATE UNIVERSITY OF ZANZIBAR SCHOOL OF NATURAL AND SOCIAL SCIENCE

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

FINAL YEAR PROJECT PROPOSAL

Project Title: IoT-Based Smart Poultry Farm

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1. Introduction

The agricultural sector remains a crucial contributor to the economy of Tanzania, with poultry farming playing a key role in food security and rural livelihoods. However, challenges such as inefficient resource utilization, poor disease management, and lack of access to modern technology hinder productivity. This project aims to address these challenges by developing an **IoT-Based Smart Poultry Farm**, integrating real-time environmental monitoring, AI-driven analytics, and smart automation to enhance farm productivity and sustainability.

2. Problem Statement

Poultry farmers in Tanzania face high operational costs, disease outbreaks, low productivity, and poor monitoring systems. Traditional methods are labor-intensive and often lack the precision and data insight needed for effective decision-making. There is an urgent need for a cost-effective, scalable, and intelligent poultry farming solution tailored for developing countries.

3. Objectives

Main Objective:

To design and implement a prototype of an IoT-based smart poultry farm management system for small to medium-scale poultry farmers.

Specific Objectives:

1. To develop an IoT system for monitoring environmental parameters (temperature, humidity, air quality, light).
2. To integrate automation systems for feeding and watering poultry.
3. To apply AI techniques for early disease detection and productivity forecasting.
4. To build a user-friendly software dashboard for farm management and analytics.
5. To evaluate the system in a simulated or actual farm environment.

4. Scope of the Project

Hardware: Temperature, humidity, gas, and light sensors; feeding/watering actuators; microcontrollers (e.g., Arduino or Raspberry Pi).

Software: Web-based dashboard, mobile notifications, cloud database.

AI Component: Disease prediction using classification models based on sensor data.

End Users: Poultry farmers, farm managers, agricultural agencies.

5. Methodology

a. **Requirements Gathering**: Through interviews and literature review of local poultry farming practices.
b. **System Design**: Architecture design combining IoT, AI, and web technologies.
c. **Hardware Development**: Integration of sensors and actuators.
d. **Software Development**: Backend (e.g., Node.js, Firebase), frontend (React/Angular), and AI model (e.g., Python/Scikit-learn).
e. **Testing & Validation**: Evaluate under lab conditions and on a local farm (if accessible).
f. **Documentation**: Prepare user manuals and final reports.

6. Expected Outcomes

- A functional prototype system for intelligent poultry farm monitoring and automation.
- AI model that provides real-time analytics and predictive alerts for disease and farm performance.
- An accessible, responsive farm management dashboard.
- Insightful evaluation data showing system performance and benefits.

7. Project Timeline

Activity	Duration	Timeline (Weeks)
Requirements Analysis	1 week	Week 1
System Design & Planning	2 weeks	Weeks 2–3
Hardware and Sensor Integration	3 weeks	Weeks 4–6
Software Development	4 weeks	Weeks 7–10
AI Model Development	2 weeks	Weeks 11–12
Testing & Refinement	2 weeks	Weeks 13–14
Documentation and Final Report	2 weeks	Weeks 15–16

8. Resources Required

i. Arduino/Raspberry Pi kits
ii. DHT11/DHT22 sensors (temperature & humidity)
iii. MQ135 or similar gas sensors
iv. Water/Feed actuators
v. Cloud hosting (Firebase, AWS)
vi. Development tools (Python, VS Code, Node.js, etc.)

9. Budget Estimate (Optional)

Item	Quantity	Unit Cost (TZS)	Total Cost (TZS)
Arduino Uno	1	50,000	50,000
Sensors (Temp, Gas, etc.)	4	15,000	60,000
Actuators	2	30,000	60,000
Other components	-	-	30,000
Miscellaneous	-	-	50,000
Total			**250,000** TZS

10. References

Guni, F. S., et al. (2021). *Performance Evaluation of Kuroiler and Sasso Chicken Breeds...*

Ngongolo, K., et al. (2021). *Chicken Production, Flock Size...*

Wilson, C. W., et al. (2022). *Diversity of Smallholder Chicken Farming...*

Ngogo, G. E., et al. (2023). *Management Systems and Productivity...*